

FUA'AMOTU, Tonga

WMO: 917920

Lat: 21.241S

Lon: 175.150W

Elev: 38

StdP: 100.87

Time Zone: 12.00 (PCF)

Period: 94-19

WBAN: 00487

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.2	16.9	11.0	8.2	19.6	12.0	8.8	19.8	9.4	22.7	8.8	22.7	0.4	150	0.327

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	5.5	30.9	26.5	30.2	26.1	29.9	25.9	27.2	29.6	26.7	29.2	26.3	28.7	4.0	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.2	21.8	28.6	26.1	21.6	28.4	25.8	21.2	28.1	86.2	30.1	84.3	29.5	82.5	28.7	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	7.9	7.1	DB	13.1	31.9	1.4	0.8	12.1	32.5	11.3	33.0	10.5	33.5	9.5	34.1	(4)
(5)			WB	12.3	27.8	1.2	0.7	11.4	28.3	10.7	28.7	10.1	29.1	9.2	29.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.2	26.4	26.9	26.5	25.5	23.9	22.8	21.8	21.7	22.2	23.2	24.4	25.6	(6)	
	DBStd	2.32	1.17	1.10	1.11	1.27	1.42	1.57	1.53	1.55	1.77	1.66	1.43	1.31	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	1	0	0	0	0	0	0	0	1	1	0	0	0	(9)	
	CDD10.0	5197	509	472	513	466	432	383	366	364	365	409	433	485	(10)	
	CDD18.3	2157	251	239	254	216	174	133	108	106	116	151	183	227	(11)	
	CDH23.3	13560	2204	2238	2237	1532	786	396	224	221	324	598	1058	1742	(12)	
(13)	CDH26.7	2449	502	577	510	246	63	18	4	6	16	49	127	331	(13)	
(14)	Wind	WSAvg	3.4	3.5	3.4	3.2	3.5	3.4	3.4	3.5	3.5	3.5	3.4	3.6	(14)	
(15) Precipitation	PrecAvg	1732	200	221	186	159	102	102	110	124	118	109	91	144	(15)	
	PrecMax	2591	532	662	413	331	317	239	235	366	371	500	263	313	(16)	
	PrecMin	928	4	42	29	18	25	17	25	16	21	24	20	11	(17)	
	PrecStd	511	163	168	93	102	83	63	62	86	88	110	69	73	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.2	31.9	31.2	30.7	29.1	28.1	27.1	27.2	28.1	29.0	29.7	30.6	(19)	
		MCWB	26.5	26.7	26.8	25.9	25.1	24.6	23.3	23.7	24.2	24.7	25.3	26.1	(20)	
	2%	DB	30.2	31.0	30.6	29.8	28.1	27.1	26.1	26.2	27.0	28.0	28.9	29.9	(21)	
		MCWB	26.1	26.6	26.4	25.8	24.6	23.7	22.7	22.8	23.4	24.2	24.8	25.6	(22)	
	5%	DB	29.9	30.2	30.0	29.0	27.2	26.2	25.2	25.2	26.0	27.0	28.0	29.1	(23)	
		MCWB	25.9	26.2	26.1	25.2	23.6	23.0	21.9	21.9	22.6	23.3	24.0	24.9	(24)	
	10%	DB	29.1	29.8	29.2	28.1	26.6	25.2	24.6	24.6	25.1	26.1	27.2	28.6	(25)	
		MCWB	25.3	26.0	25.5	24.5	23.1	22.2	21.4	21.3	21.7	22.5	23.4	24.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.6	27.7	27.7	26.8	26.2	25.6	24.6	24.7	25.1	25.6	26.2	26.8	(27)	
		MCDB	29.8	29.9	30.0	29.0	28.1	27.2	26.1	26.4	27.1	27.9	28.4	29.3	(28)	
	2%	WB	26.9	27.2	27.0	26.2	25.5	24.7	23.8	23.8	24.1	24.8	25.6	26.2	(29)	
		MCDB	29.3	29.7	29.5	28.5	27.3	26.3	25.2	25.5	26.0	27.0	27.7	28.8	(30)	
	5%	WB	26.4	26.7	26.5	25.8	24.8	24.0	23.1	23.0	23.5	24.1	25.0	25.7	(31)	
		MCDB	28.7	29.2	28.9	28.0	26.4	25.6	24.6	24.7	25.3	26.1	27.1	28.2	(32)	
	10%	WB	26.0	26.2	26.1	25.3	24.2	23.2	22.3	22.2	22.7	23.5	24.5	25.3	(33)	
		MCDB	28.3	28.6	28.3	27.4	25.8	24.8	24.0	24.0	24.5	25.4	26.5	27.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.4	5.5	5.4	5.2	5.2	5.2	5.4	5.4	5.6	5.7	5.6	5.6	(35)	
		MCDBR	6.2	6.2	6.2	6.1	5.9	6.1	6.3	6.2	6.4	6.3	6.2	6.1	(36)	
	5% WB	MCWBR	3.2	3.2	3.3	3.4	3.5	3.7	3.9	3.9	3.9	3.6	3.3	3.3	(37)	
		MCWBR	5.5	5.8	5.5	5.6	5.4	5.6	5.3	5.8	5.6	5.7	5.5	5.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.384	0.376	0.378	0.390	0.371	0.369	0.358	0.361	0.381	0.369	0.371	0.379	(40)	
		taud	2.611	2.645	2.645	2.588	2.619	2.607	2.626	2.618	2.551	2.583	2.601	2.597	(41)	
	Ebn at Noon		961	956	929	877	856	838	862	893	912	952	966	966	(42)	
		Edh at Noon	104	98	95	93	84	82	82	89	102	104	104	105	(43)	
(44) All-Sky Solar Radiation	RadAvg		6.38	6.15	5.45	4.59	3.88	3.42	3.69	4.28	5.11	5.96	6.45	6.71	(44)	
	RadStd		0.58	0.54	0.38	0.51	0.25	0.23	0.22	0.38	0.40	0.41	0.50	0.38	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	+1.04	N/A	+0.71	+0.78	N/A	N/A	N/A	N/A	(46)
(47)	Regional (1 neighbor)	N/A	N/A	+1.04	N/A	+0.71	+0.78	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page